

Question

Recently, OL published an article on methodology for the synthesis of a 6-5-5 tricyclic system. The reaction scheme is shown in Figure 1, where the orientation of the hydrogen atom marked with * is unknown.

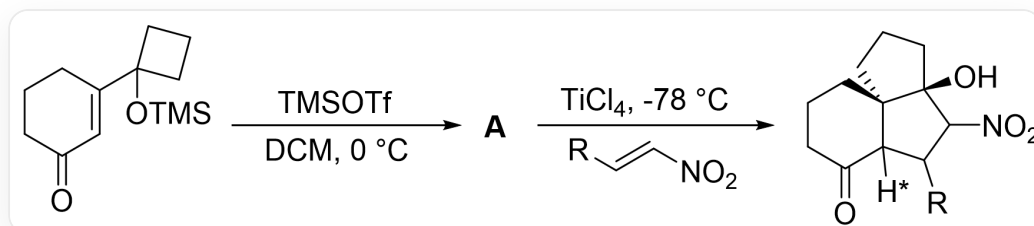


Fig. 1, the figure shows a two-step sequential reaction. The first step reaction is represented by SMILES as:

O=C1CCCC(C2(O[Si](C)(C)C)CCC2)=C1>>[A], the reaction conditions are TMSOTf, DCM, 0°C. The

second step reaction is represented by SMILES as: [A]>[R]/C=C/[N+](O)=O>O[C@]1(C(C(C23[H*]))[R])

[N+](O)=O)CCC[C@@]31CCCC2=O. The reaction conditions are TiCl4, -78°C

Deduce the structure of **A**, considering the orientation of the hydrogen atom marked with *.

There are the following statements:

1. **A** contains three rings of six members or less.
2. **A** contains a carbonyl group located on a six-membered ring.
3. With the five-membered ring containing **R** as the plane, the reaction product with **H*** pointing out of the plane of the paper is more abundant.
4. With the five-membered ring containing **R** as the plane, the reaction product with **H*** pointing into the plane of the paper is more abundant.

Among the following options, the one with all correct statements and the largest number of correct statements is:

A. All other options are incorrect

B. 1

C. 2

D. 3

E. 4

F. 1,2

G. 1,3

H. 1,4

I. 2,3

J. 2,4

K. 1,2,3

L. 1,2,4

Answer

Correct Answer: **D**

Detailed Explanation

The first step, under the action of a strong Lewis acid, is a Pinacol-like rearrangement reaction, which opens the unstable four-membered ring, forming a more stable five-membered spirocycle, forming intermediate **A** as shown in Figure 2.

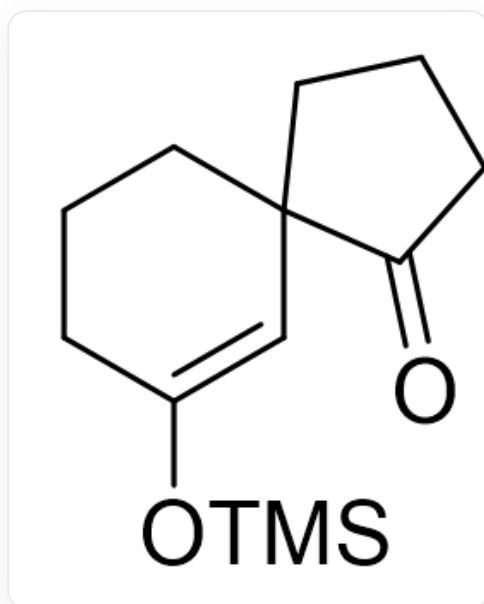


Fig. 2, the molecule is represented by SMILES as: O=C(CCC1)C21C=C(O[Si](C)(C)C)CCC2

CHECKPOINT

1 PTS

Pinacol-like rearrangement occurs to form the intermediate O=C(CCC1)C21C=C(O[Si](C)(C)C)CCC2

A contains only two rings, statement 1 is incorrect. The carbonyl group is located on the five-membered ring, statement 2 is incorrect.

After adding the unsaturated nitroalkene, a Michael addition reaction occurs, yielding a carbanion, which further undergoes an addition reaction with the nearby carbonyl group, and workup yields the final product.

CHECKPOINT

1 PTS

The silyl enol ether further undergoes a Michael addition reaction, and the resulting carbanion is coupled with the nearby carbonyl group to obtain the product given in the question

When **A** is added to the nitroalkene, the silyl enol ether can initiate an attack from above the six-membered ring, i.e., near the carbonyl group of the five-membered spirocycle, or from below the six-membered ring, i.e., near the methylene group of the five-membered spirocycle. Both the intramolecular carbonyl group and the enol hydroxyl group are bonded to the Lewis acid TiCl_4 . The carbonyl group has a larger steric hindrance, while the methylene group contains only two hydrogen atoms and has a smaller steric hindrance. Therefore, the reaction preferentially occurs on the side away from the carbonyl group, resulting in a reaction product with the five-membered ring where **R** is located as the plane and **H**^{*} pointing out of the paper. Statement 3 is correct, and statement 4 is incorrect.

CHECKPOINT

1 PTS

Since the carbonyl group is bonded to the Lewis acid TiCl_4 and has a larger steric hindrance than the methylene group, the Michael addition reaction preferentially occurs on the side away from the carbonyl group